Capital Allocation II

Factor bets and alpha bets

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Today

In the slides

- Factor bets against alpha bets
- The same formula, on hedged portfolios
- Putting a number on a hunch

In the notebook, you will

- Size alpha bets on five factors and combine them with the market
- Run all 29 strategies: the simple rule against the whole matrix
- Put in your own views: a taste of Black-Litterman

Homework: cook up four trading ideas, calibrate them and size the book

Last class was about factor bets. Today adds alpha bets.

	a factor bet	**an alpha bet**
what it is	exposure anyone can buy: the market, value, Tech	what is left after you hedge those
what it interacts with	your job, your house, your mandate, your values	nothing you own, if it is truly alpha
follow the formula?	loosely: tilt for all of that	yes: maximize the Sharpe ratio

Every tilt from last class lives in factor space. True alpha needs no tilt. The catch is the word *true*.

The same engineer, two trades

	long Tech	**long one chipmaker, short another, hedged**
what it adds	more of what the job already holds	something the job does not move with
how you size it	less than the formula says, even with a high μ	by the formula: appraisal ratio
what you must believe	a premium, earned for bearing risk	the market is wrong, and you know why

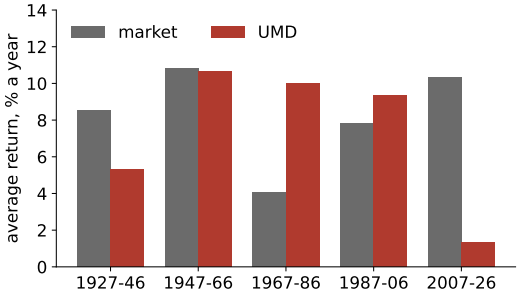
- The more you trust an alpha, the less you need any other consideration
- So the hard question moves from "how much do I want?" to "is it real?"

Two different objects, so two different estimates

	a factor's mean	**an alpha**
where it comes from	pay for bearing risk	mispricing, or an edge in information
how long it lasts	as long as the risk does: slow	until others find it: it decays
which data	all you have, the longer the better	the recent past, and often not returns at all
the prior	positive, from economics	zero, until shown otherwise
the horizon	years	can be one event, then gone

- The covariance is the same for both: recent data, high frequency, and it learns fast

What the data say about persistence



The market: 4% to 11% in every 20 years. But UMD, once published and crowded: 9% to 1%. A style factor can behave like an alpha.

49 industries	5 years on next 5
correlation of α	-0.12
correlation of β	$+0.61$
correlation of vol	$+0.75$

- Our 29: alpha correlates 0.78 across 1980–2000. They are on the list because they worked.
- After publication, such returns fall 58% (McLean and Pontiff 2016)

Your strategy is an alpha bet and a factor bet

$$r_i \;=\; \alpha_i \;+\; \beta_i\, f \;+\; \varepsilon_i$$

r_i: your strategy $\qquad$ f: the factor, here the market $\qquad$ ε_i: what is left

Hedge out the factor and trade what is left:

$$h_i \;=\; r_i - \beta_i\, f \qquad \text{mean } \alpha_i, \quad \text{volatility } \sigma_{\varepsilon,i}, \quad \text{no market exposure}$$

- Anyone can buy the market for free. The hedged piece is what you are sizing here.
- So the assets become the h_i, the means become α_i, and the covariance becomes Σ_ε

Where the inverse disappears

If the residuals are uncorrelated, Σ_ε is diagonal, and inverting it just divides:

$$W^\star \propto \Sigma_\varepsilon^{-1}\alpha \qquad \Longrightarrow \qquad w_i \propto \frac{\alpha_i}{\sigma^2_{\varepsilon,i}}$$

Multiply by $\sigma_{\varepsilon,i}$ to get the volatility you give strategy i:

$$\underbrace{w_i\,\sigma_{\varepsilon,i}}_{\text{volatility you give it}} \propto \underbrace{\frac{\alpha_i}{\sigma_{\varepsilon,i}}}_{\text{its appraisal ratio}}$$

- No matrix. The number you have computed since *Performance Evaluation* is a position size.
- If they are correlated it is the same formula, with the whole matrix: $W^\star \propto \Sigma_\varepsilon^{-1}\alpha$

Bigger alpha does not mean a bigger position

	α	σ_ε	appraisal ratio	weight $\alpha/\sigma_\varepsilon^2$	volatility you give it
Strategy A	4%	4%	1.0	25	1.0
Strategy B	8%	16%	0.5	3	0.5

- B has twice the alpha and gets half the risk, and an eighth of the weight
- The formula divides by a square, so it loves whatever looks quiet

What is combining worth?

You run two strategies. Each has a Sharpe ratio of 1.0.
Their correlation is zero. You size them optimally.

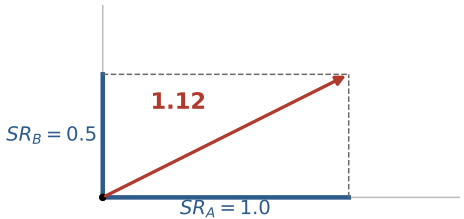
What is the Sharpe ratio of the two together?

A 1.0

B 1.41

C 2.0

Correlation is an angle



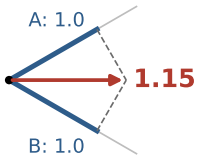
- Draw each strategy as a direction. The angle between two directions has cos(angle) $= \rho$.
- The best combination is a vector, in red. Its length is the combined Sharpe ratio.
- Each strategy's Sharpe ratio is the shadow of that vector on its direction
- Uncorrelated means a right angle, so Pythagoras: $\sqrt{1.0^2 + 0.5^2} = 1.12$

Squared Sharpe ratios add *because* the bets are at right angles

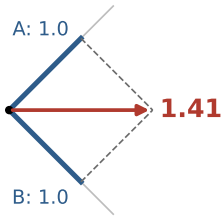
Why

Change the angle: two strategies, each with a Sharpe ratio of 1

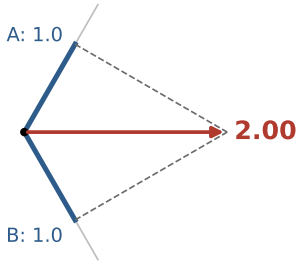
$\rho = +0.5$, angle 60°



$\rho = +0.0$, angle 90°



$\rho = -0.5$, angle 120°



- **Positive correlation, under 90°:** they overlap: 1.15, less than Pythagoras.
- **Negative correlation, over 90°:** each one hedges the other: 2.00, more than Pythagoras.

Judge a strategy by what it adds to what you hold, not by its Sharpe ratio alone

The formula needs two inputs. You treat them differently.

Σ_ε	**estimate it**	recent data, high frequency
α	**a track record**	$\hat{\alpha}$, with a standard error
	a signal	your signal's forecast, scaled
	a view	a hunch, made into a number

- In practice most of the thinking goes into α and most of the data goes into Σ_ε
- The formula cannot tell which α came from where. You have to.

Estimate the risk. Supply the alpha.

A hunch is not an input until it is a number

- "I like Apple." Compared to what, by how much, by when, and how sure are you?
- Leaving it out is also a number. It is $\alpha = 0$, and the formula will act on it.
- Once it is a number it can be sized, combined with your other bets, and scored when the date arrives
- Without a number you cannot be wrong, and so you cannot learn. That was *Hiring and Firing*, applied to yourself.

The discipline is to write the number down before the event

From a hunch to a number: Apple's earnings will beat consensus

the question	your answer (stylized)	the number
What is in the price?	consensus earnings \$1.60; you say \$1.68	a 5% beat
What is a beat worth?	past 5% beats moved the stock about 4%: estimate it	$\pm 4\%$
How sure are you?	right 60% of the time; wrong means a miss	$(0.6 - 0.4) \times 4\%$
By when?	the report is in six weeks, then the edge is gone	30 trading days
How much risk?	25% a year of stock-specific volatility: $25\%\sqrt{30/252}$	8.6%

$\alpha = +0.8\%$ over six weeks, on Apple hedged for the market and its industry

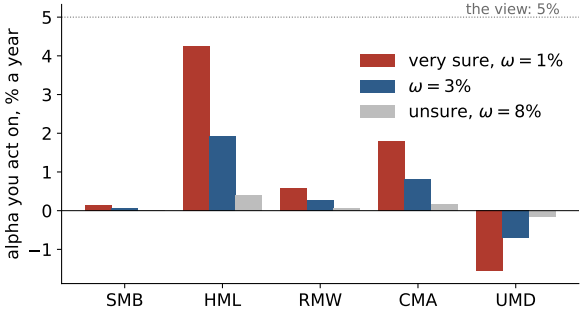
- It is not 6.7% a year. You earn that only if you have another idea like it every six weeks.

Check the number: what have you just claimed?

	right 60%	right 55%
α over six weeks	0.8%	0.4%
α/σ over six weeks: the IC you are claiming	0.09	0.05
appraisal ratio, a year	0.27	0.13
position, in a 10%-volatility book with Sharpe ratio 1	11%	5%

- Good managers run an IC of about 0.02 to 0.05 (Grinold and Kahn). Is this the best idea you have ever had?
- Position: volatility given $= (AR/SR_{\text{book}}) \times 10\%$, so the weight is that over 25%
- Being honest about your confidence halved the position. That is the formula working.

A taste of Black–Litterman: start from no alpha, add your views



$$\alpha_{\text{act}} = \frac{\tau^2}{\tau^2 + \omega^2}\,\alpha_{\text{view}}$$

τ: how big alphas tend to be.

ω: your view's standard error.

- One view: HML's alpha is 5%
- Less sure, smaller bet
- Its neighbours' alphas move too, CMA up and UMD down, and that is why you end up holding only HML

Your alpha is an estimate, so shrink it. That is *Capital Allocation III* in one line.

The whole formula

Now: the notebook

Notebook: Capital Allocation II — the same formula, on hedged portfolios

1. **Alpha bets on five factors.** Hedge, size, build the book, combine it with the market.
2. **All 29 strategies.** The rule against the whole matrix. Look at the positions.
3. **Your views.** Put one number in, and watch the book move.

Bring back: the Sharpe ratio of your book, of the market, and of the two together. Does the formula hold?

Stop at **Back to the slides**; the challenge after it is homework.

0.57 and 0.97 make 1.12

	Sharpe ratio, 1980–2026
the market	0.57
the book: five factors, hedged, sized by the rule	0.97
the two together	1.12
$\sqrt{0.57^2 + 0.97^2}$	1.12
the book, if its five pieces were uncorrelated	1.10

- The hedge makes the book uncorrelated with the market, so the formula is exact
- Inside the book it falls short: 0.97, not 1.10. HML and CMA are correlated 0.66.

So: how much does correlation cost when you have many bets?

Why pod shops exist

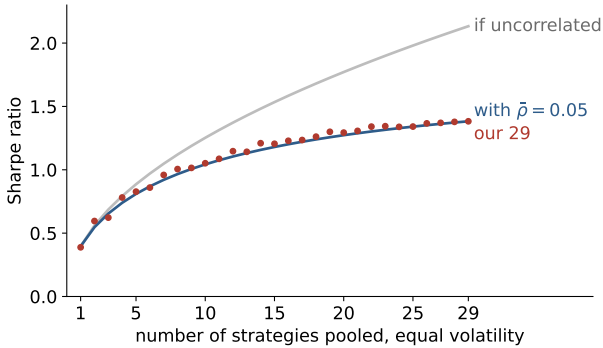
$$SR_{\text{pool}} = \sqrt{N} \times SR_{\text{pod}}$$

N uncorrelated teams, each hedged, each with the same Sharpe ratio. The last formula, with N terms.

- A pod with a Sharpe ratio of 0.5 is not a fund. Nobody allocates to it.
- Thirty of them, uncorrelated, make 2.7. That is one of the best records in the industry.
- Citadel and Millennium run many small hedged teams, each on its own risk budget

The pods are worth more together than apart, and that gap is the firm

Our 29 strategies: 0.40 apart, 1.38 together



- $\sqrt{29} \times 0.40 = 2.13$. We get 1.38.
- The whole gap is correlation:

$$\bar{s}\sqrt{\frac{N}{1+(N-1)\bar{\rho}}} = 1.38$$

- $\bar{\rho}$ is only 0.05. Even with endless strategies you stop at 1.78.

Why

The scarce thing is not a good strategy. It is an uncorrelated one.

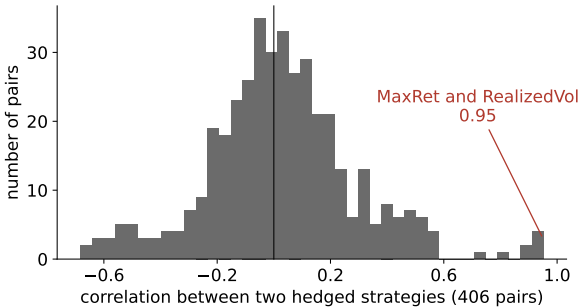
One result, three forms: skill times breadth

two uncorrelated bets	$SR = \sqrt{SR_A^2 + SR_B^2}$
N pods	$SR_{\text{pool}} = \sqrt{N}\, SR_{\text{pod}}$
the Fundamental Law (Grinold 1989)	$IR = IC \times \sqrt{\text{breadth}}$

IC: the correlation between your forecasts and what happened. Breadth: independent bets a year.

- $IC = 0.03$ is barely detectable. With 1,000 independent bets a year, $IR \approx 0.95$.
- It names the pattern you just derived twice, and it is the IC you claimed for Apple.

The catch



- Average correlation 0.03. It looks like zero.
- Average *absolute* correlation 0.20
- Some pairs are one trade under two names

Every formula so far assumed you know μ and Σ. You estimated them.
Capital Allocation III: what that costs.

Now you: four ideas, before you look at any data

Using the 49 industries, each idea is a hedged long-short. Discuss, then write them down:

- **What do you think you know?** AI spending, interest rates, oil, an aging population, a recession.
- **Why is it not in the price?** If you cannot say, your alpha is zero.
- **By when, and how sure?** Then one number for each.
- **Are they four ideas?** Two that are correlated 0.9 are one idea.

Homework: four trading ideas. Write the number first. Then estimate the risk, look at the record, shrink, and size the book. The memo defends your biggest position.

Takeaways

1. Factor bets interact with your life, so tilt them. True alpha does not, so maximize the Sharpe ratio.
2. A factor's mean is slow and an alpha decays. Long history for one, signals and judgment for the other.
3. It is the same formula on hedged portfolios. Without correlation, volatility follows the appraisal ratio.
4. Correlation is an angle: under 90° adds less than Pythagoras, over 90° adds more.
5. Estimate the risk, supply the alpha, and write the number down before the event.
6. Breadth is the prize, and correlation limits it: 0.40 apart, 1.38 together, not 2.13.

Appendix

A: Why the picture is exact

Two strategies with Sharpe ratios s_1, s_2 and correlation ρ. The best combination has

$$SR^2 = \frac{s_1^2 + s_2^2 - 2\rho\, s_1 s_2}{1-\rho^2}$$

- Any asset's Sharpe ratio equals the best Sharpe ratio times its correlation with the best portfolio: $s_i = SR\,\rho_{i,\star}$
- That is the shadow: $SR\cos(\text{angle})$. The red vector is the one whose two shadows are s_1 and s_2.
- With $s_1 = s_2 = s$: $SR = s\sqrt{2/(1+\rho)}$, so $1.15s$, $1.41s$, $2s$ at $\rho = 0.5, 0, -0.5$
- If $s_2/s_1 < \rho$, the red vector falls outside the two directions: a short position, as in *Capital Allocation III*

Back

A: Black–Litterman, the whole formula

$$\alpha_{\text{act}} \;=\; \pi \;+\; \tau\Sigma P' \left(P\,\tau\Sigma\,P' + \Omega\right)^{-1}\left(Q - P\pi\right)$$

π: the prior; for hedged bets, zero. $\tau\Sigma$: the prior's covariance; here $\tau = 0.05$.
P: which portfolios your views are on. Q: the numbers. Ω: how unsure you are.

- One view on one asset: P picks it out, and the first term becomes $\tau^2/(\tau^2 + \omega^2)$ with $\tau^2 = \tau\sigma^2$
- The Σ in front moves every asset correlated with your view. Then $\Sigma^{-1}\alpha_{\text{act}}$ cancels it: with $\pi = 0$ the weights are a multiple of P', so you hold only what you have a view on
- In the original, π comes from the market's weights, so with no views you hold the market (Black and Litterman 1992)

Back

A: Why correlation sets the limit

Scale N strategies to the same volatility σ, and hold them equally. Let $\bar{s}$ be their average Sharpe ratio and $\bar{\rho}$ their average pairwise correlation.

$$\text{mean of the pool} = \bar{s}\,\sigma \qquad \text{variance of the pool} = \frac{\sigma^2}{N}\Big(1 + (N-1)\bar{\rho}\Big)$$

$$SR_{\text{pool}} \;=\; \bar{s}\sqrt{\frac{N}{1+(N-1)\bar{\rho}}} \;\xrightarrow{N\to\infty}\; \frac{\bar{s}}{\sqrt{\bar{\rho}}}$$

- This is an identity, not an approximation, which is why it matches the 29 to three decimals
- $\bar{\rho} = 0$ gives back $\sqrt{N}\,\bar{s}$
- In the figure, each red dot averages 300 random sets of N strategies

Back

A: Data, and what to be careful about

- **Six factors.** Ken French: market, SMB, HML, RMW, CMA, UMD. Monthly, 1980–2026, 558 months. UMD and the 49 industries go back to 1927.
- **The 29 strategies.** Long-shorts on published signals, on our CRSP panel, 1980–2000, 251 months.
- **Selection.** These signals are in the literature because they worked, much of it in this sample. Their Sharpe ratios are overstated as forecasts of what comes next.
- **In sample.** Alphas, betas and residual volatilities use the full period. Every Sharpe ratio built from them is in sample, and so is every book.
- **The Apple numbers are stylized.** Estimate the response to a surprise before you use it.